WENXUAN LIU

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EDUCATION

- **Columbia University**, New York, NY — Ph.D. Candidate in Operations Research (advised by Adam N. Elmachtoub, Tianyi Lin), 2024-Present, GPA 4.05/4.3
- **Tsinghua University**, Beijing, China — B.S. in Industrial Engineering (Minor: Software Engineering) 2019-2024, GPA: 3.98/4.0 (Rank 1/67)

PUBLICATION AND PREPRINTS

1. Adam N. Elmachtoub, Tianyi Lin, Wenxuan Liu(α-β), Preventing Model Extraction Attacks in Contextual Optimization, working paper.
2. Wenxuan Liu, X. Liu, M. Liu, H. Qin, Z. Zhang. Tackling Decision Dependency in Contextual Stochastic Optimization. Production and Operations Management, 2026.

RESEARCH EXPERIENCES

1. Adam N. Elmachtoub, Tianyi Lin, Wenxuan Liu(α-β), Preventing Model Extraction Attacks in Contextual Optimization. Working paper.

   Proposed the first attack and defense framework for model extraction in contextual optimization (predict-then-optimize) systems. Established the theoretical foundations of model vulnerability and protection against extraction attacks.

2. Wenxuan Liu, X. Liu, M. Liu, H. Qin, Z. Zhang. Tackling Decision Dependency in Contextual Stochastic Optimization. Production and Operations Management, 2026.

   Studied contextual stochastic optimization with decision-dependent uncertainty. Proposed the contextual gradient and the corresponding contextual gradient-based algorithm with convergence guarantee to address the challenge of decision dependency.

3. Wenxuan Liu, R. Zhu and H. Qin, Dynamic Pricing with Price Protection Guarantee. Summer Research Project.

   Investigated online learning algorithms for dynamic pricing under "price protection" guarantees (refunds to earlier buyers if price drops). Developed online algorithms with near-optimal regret bounds for single-product pricing and extended the analysis to multi-product cases.

TALKS

- Tackling Decision Dependency in Contextual Stochastic Optimization. *2025 Purdue Supply Chain & Operations Management Conference, (August, 2025)*

INDUSTRY EXPERIENCE

Qingzhi Optimization, Tianjin, China, *Optimization Engineer Intern* 2021-2022

Bytedance, Bellevue, WA, *Machine Learning Engineer Intern* Summer 2026

SKILLS AND AWARDS

Skills: Python; C/C++; (basic) Java, R, MATLAB. Proficient in machine learning, prescriptive analytics, and optimization (linear/integer programming, dynamic programming).

Awards: Columbia University IEOR Fellowship (2024); Outstanding Graduate of Beijing (2024); Academic Excellence & Innovation Scholarships, Tsinghua University (2022-2023); Comprehensive Excellence Scholarship, Tsinghua University (2020-2021).